

Riddhesh Dalal

Email: riddhu.work@gmail.com

Phone: +91-8982094147

EDUCATION

GitHub: https://github.com/artignite3

LinkedIn: https://www.linkedin.com/in/riddhesh-dalal

Indian Institute of Technology, Patna

B. Tech, Chemical Science and Technology

Macro Vision Academy, Burhanpur

Senior Secondary (XII) Board Examination (CBSE)

Macro Vision Academy, Burhanpur

Secondary (X) Board Examination (CBSE)

CGPA: 8.04

July 2024 - Present

Percentage: 88.2%

June 2022 - May 2024

Percentage: 97.2%

July 2013 - May 2022

EXPERIENCE

Undergraduate Research Assistant, Drisha Lab | Prof. Sriparna Saha

Apr. 2026 – Present

Machine Learning, Deep Learning & Generative AI

- Proposed Persistent Tactical Dynamics (PTD), a novel framework for modeling evolving team tactics using Hidden Markov Models and spatio-temporal player interactions.
- Implemented baseline models on SoccerNet-v3 benchmark (500 matches, 764 hours), integrating multi-object tracking with spatio-temporal graph neural networks for tactical impact analysis and game-state value modeling.
- Architected SM3 (TactiGraph: Tactical Player-Contribution), a player-contribution modeling system on SoccerNet video data using spatial Gaussian fields and pressure-sensitive ball-carrier metrics; research work currently being prepared for A/top-tier conference submission

Sub-Coordinator - Inter IIT Tech 14.0

Sept. 2025 - Dec. 2025

Official Website & Security Application, Inter-IIT Tech Meet 14.0

Source Code

- **Tools & Technologies:** Next.js, React.js, Node.js, MongoDB, TypeScript, JWT, Vercel, React Native
- Developed and deployed the official full-stack website featuring multiple problem-statement, admin/contingent registration, and data-driven dashboards serving 23 IITs and supporting 2000+ contingent registration; hardened backend security with JWT authentication, and encrypted databases, Integrated a React Native companion app for real-time participant entry/exit management during the presentation.

PROJECTS

Redrob Intelligent Candidate Discovery Engine (Team Lead, Team MM – 4 members)

June 2026 - July 2026

Python, BM25, FAISS, Sentence-Transformers, Cross-Encoder, Qwen2.5-LLM

Source Code

- Built an explainable multi-stage AI candidate-ranking pipeline, ranking 100,000 candidates pool entirely CPU within a 5 minutes time limit for a hackathon.
- Designed a 7-stage AI candidate discovery pipeline, integrating BM25 lexical retrieval, embeddings(BGE-base-en-v1.5) with FAISS semantic search, behavioral signals, and neural re-ranking and natural-language rank explanations via Qwen 2.5-1.5B-Instruct
- Rectified a critical scoring bug by identifying an unflattened nested dictionary, restoring differentiated rankings across 100,000 candidate records.

AI-Driven Gamified Team Predictor - New X1

Dec. 2025 - June 2026

Python, XGBoost, CatBoost, PuLP, FastAPI, Streamlit, SHAP, Ollama/Llama

Source Code

- Engineered end-to-end ML pipeline predicting individual cricket player scores across 3000+ matches, 35000+ players records with ball-by-ball records and match conditions, then auto-generated optimal 11-player squads respecting role and composition limits.
- Benchmarked 2 Gradient boosting models, **XGBoost** and **CatBoost** regressors on 140 synthetic features for fantasy-point prediction; applied **PuLP** integer programming for team-selection optimization, with **SHAP**-based per-player explainability surfaced through a **FastAPI** + **Streamlit** application exposing real-time prediction APIs

Faculty Recruitment Portal (Collaborative – Team of 2, with IIT Ropar)

Nov. 2025 - Feb. 2026

Node.js, MySQL, EJS, Bootstrap, JavaScript

Source Code

- Delivered a full-stack recruitment portal with session-based authentication and role-based access for applicants/admins, backed by a normalized **MySQL** schema managing qualifications, employment history, and publications.

SKILLS/RELEVANT COURSEWORK

- **Languages:** Python, C/C++, JavaScript, TypeScript, SQL, HTML/CSS
- **ML/AI:** PyTorch, XGBoost, CatBoost, SHAP, Graph Neural Networks, Ollama/Llama, LangChain, LLM APIs
- **Web & Backend:** Node.js, FastAPI, Next.js, React.js, REST APIs, JWT
- **Databases & Storage:** MySQL, MongoDB, DuckDB
- **Tools & Platforms:** Git, GitHub, Linux, Vercel, Streamlit, React Native (Basic)
- **Key Courses:** Data Structures & Algorithms, Intro to Data Science, Probability & Statistics

ACHIEVEMENTS

- **300+** DSA Problems and **100+** SQL Queries solved (June 2026) • **Rank 13,641** (Top 5%) among 0.25 million candidates in JEE Advanced (May 2024) • **99.4%ile** – JEE Mains, out of 1.35 million candidates (Jan 2024) • **MAS, Top 1% (M.P.)** – IAPT NSEP Olympiad (Nov. 2023) • **Zonal Rank 1** – NSO, IMO (SOF Olympiad) (Mar. 2022)

POSITIONS OF RESPONSIBILITY

- **Overall Coordinator**, Vincetroke - Fine Arts Club, IIT Patna (Apr. 2026 – Present) • **Production Lead**, TEDx IIT Patna (Apr. 2026 – Present) • **Contingent Member**, Inter-IIT Cult 8.0 (Dec. 2025) • **Dy. Cultural Minister**, Class 11th (June 2022) • **Head Boy**, Class 10th (Apr. 2021)